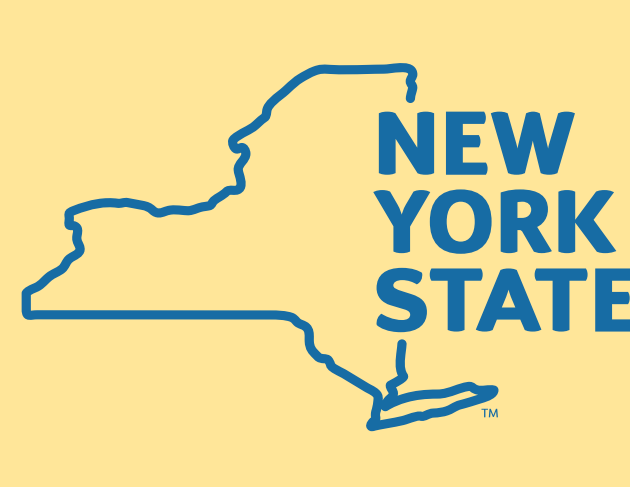
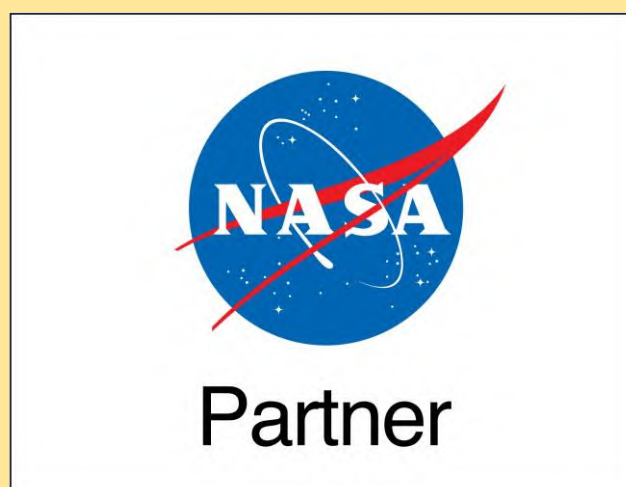


# GEOS-Chem-APM for (1) physics-guided machine learning parameterizations & (2) aerosol pollution exposure and health disparities assessment



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Funding:

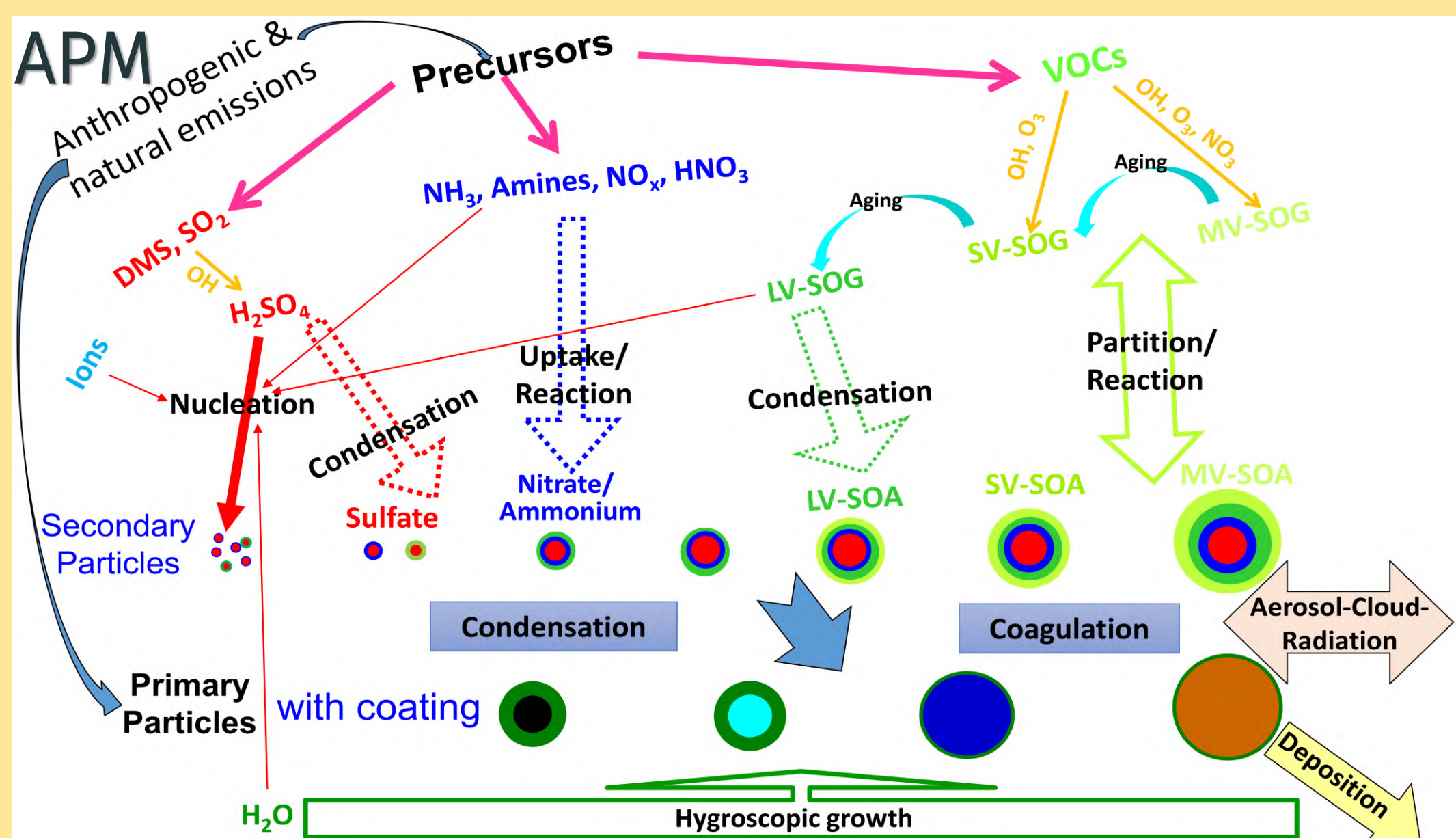


NYSDA  
Supported

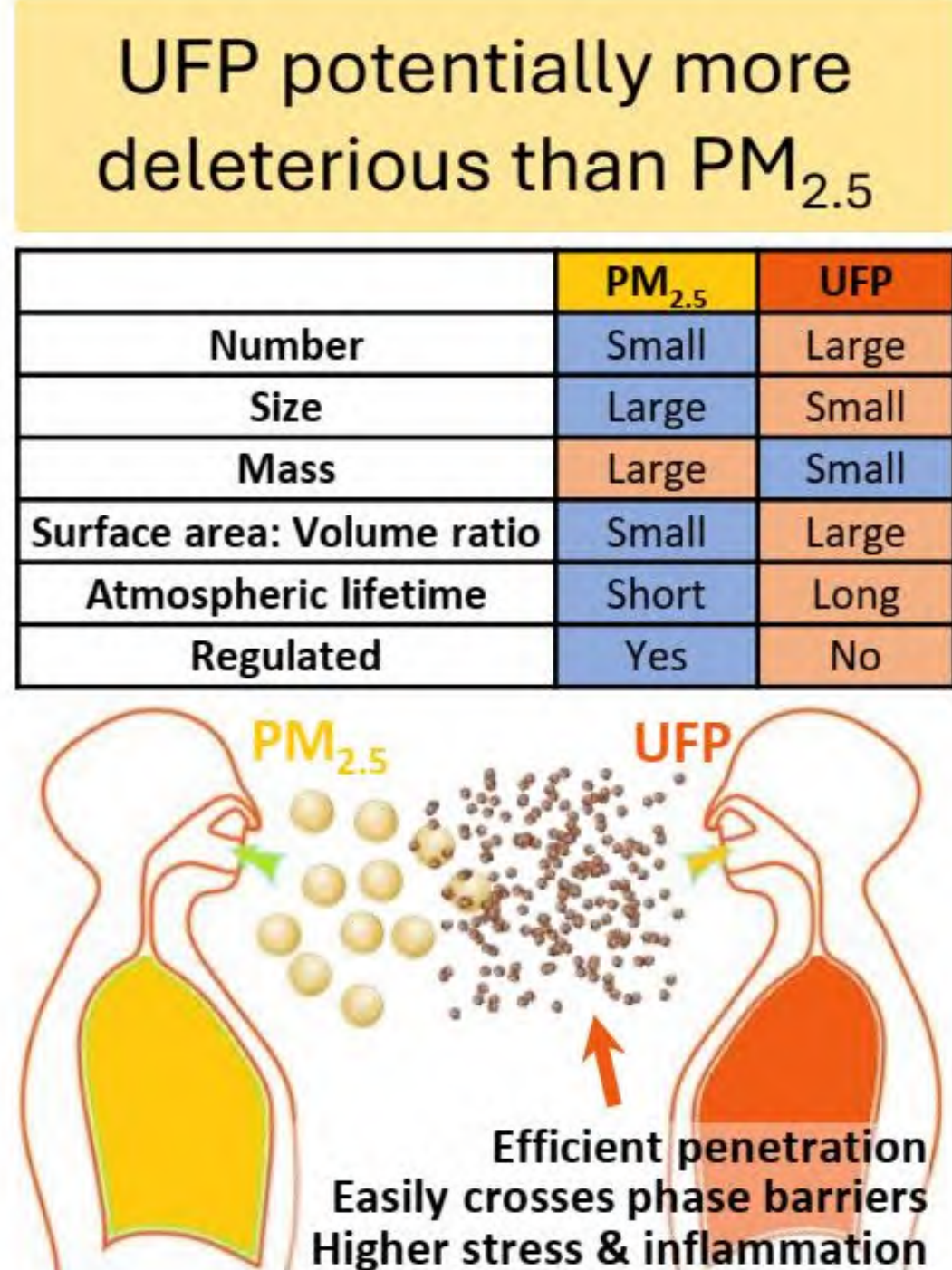
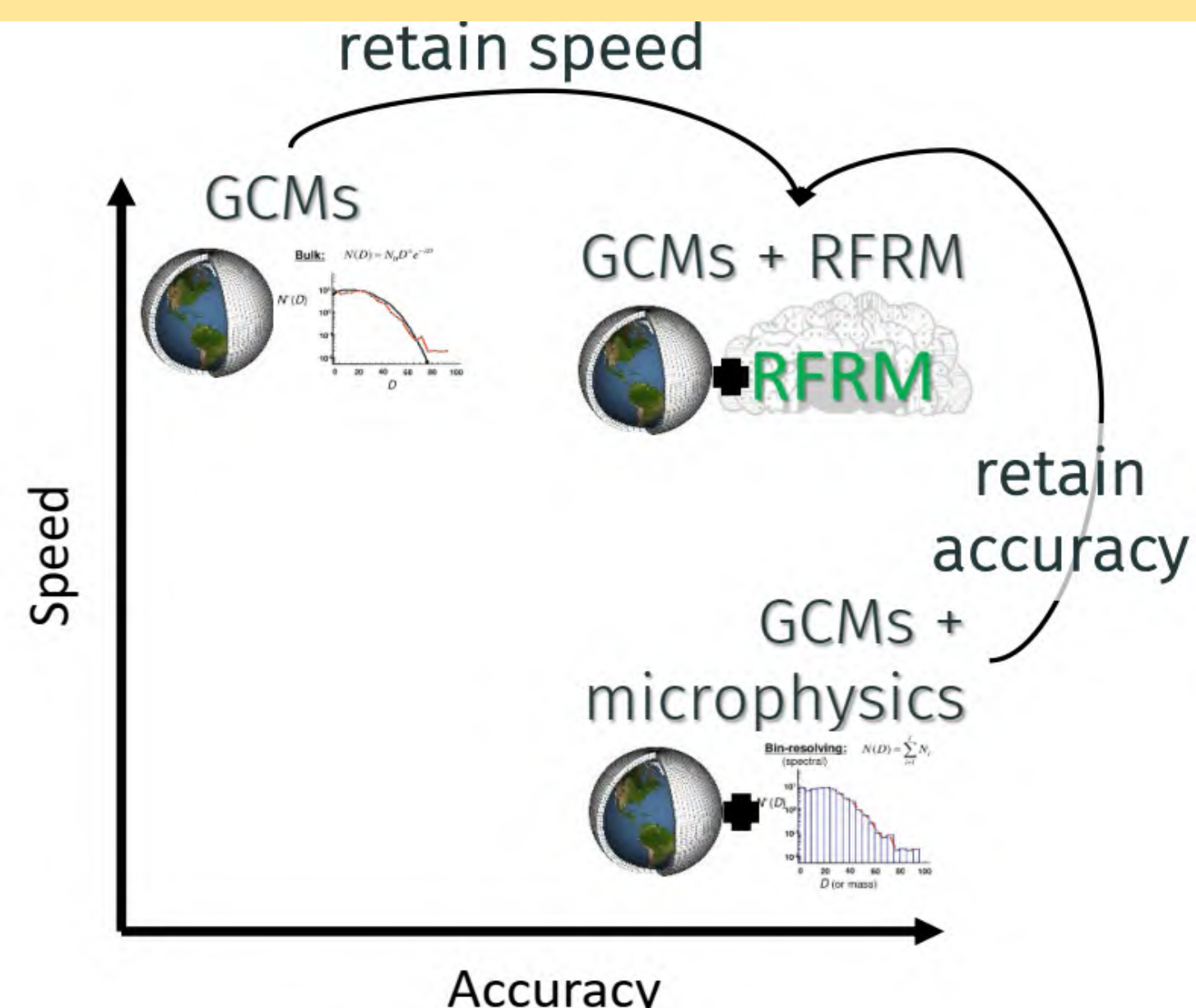


## Research Tool

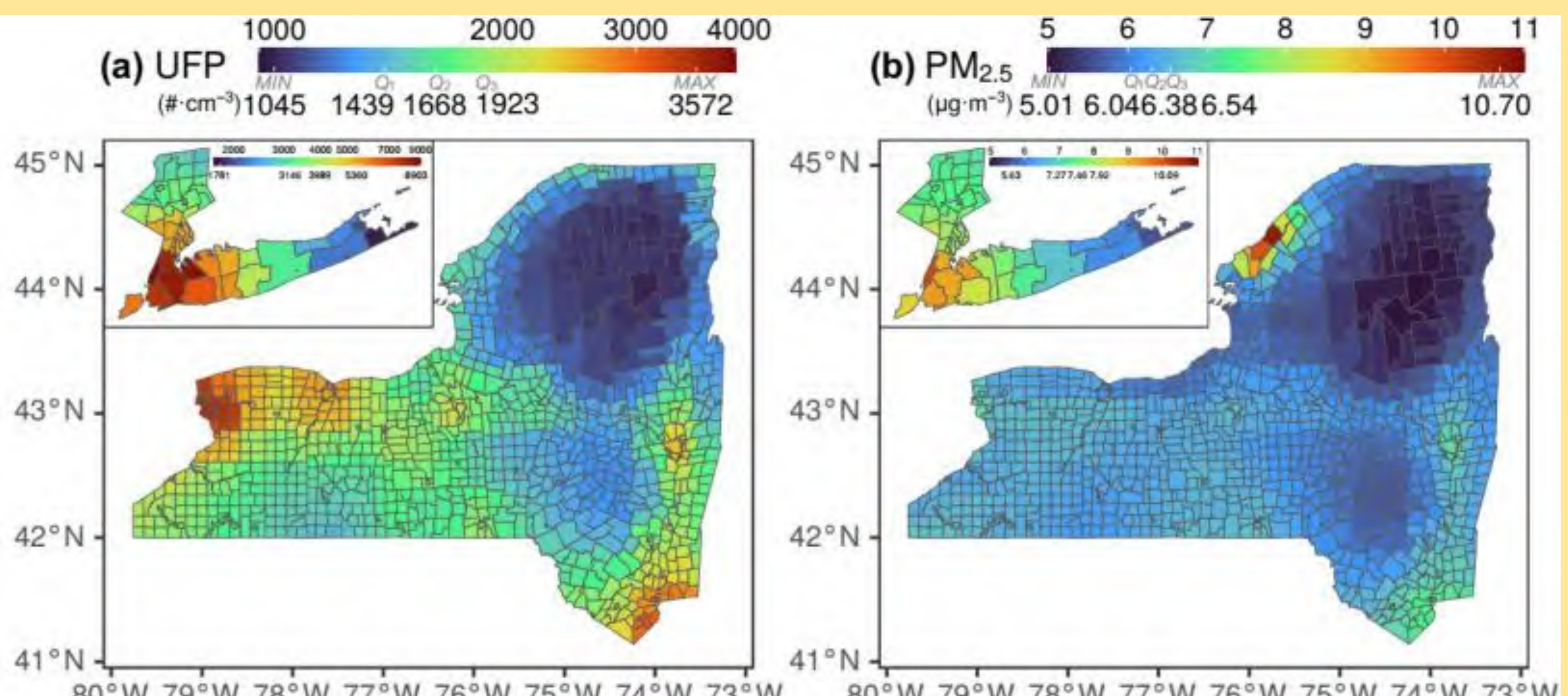
GEOS-Chem-APM  
Full Chemistry  
Size-resolved (bin) particle microphysics  
Coating of primary particles by secondary particles  
State-of-the-science nucleation mechanisms  
(Yu et al., GMD 2020)



## Objectives



## Domain for health-effects studies

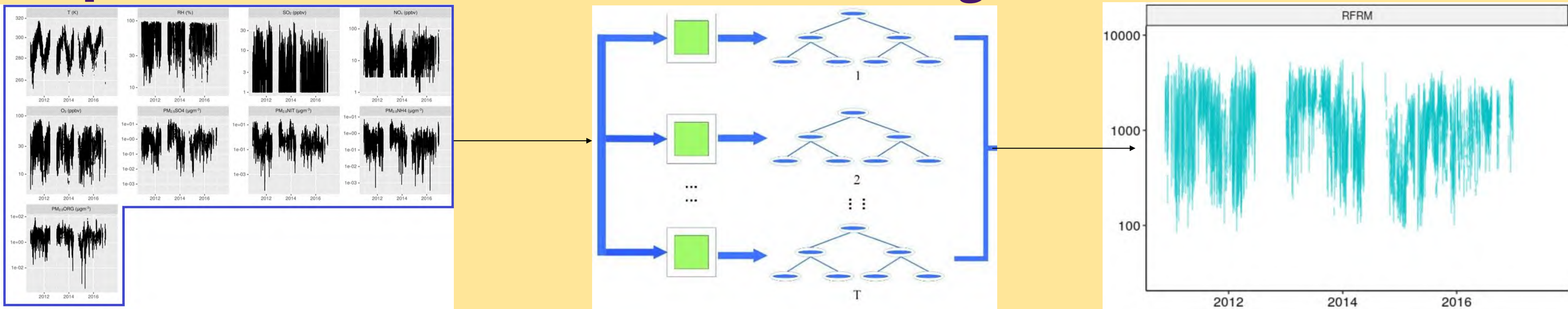


## Machine Learning of Aerosol Properties

### Input variables

### Machine learning

### Predictions



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## Using machine learning to derive cloud condensation nuclei number concentrations from commonly available measurements

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- ML trained on long-term GEOS-Chem-APM simulations
- Predictors: fractions of PM<sub>2.5</sub> (NH<sub>4</sub>, SO<sub>4</sub>, NO<sub>3</sub>, SOA, BC, POC, dust, and salt), gaseous species (NO<sub>x</sub>, NH<sub>3</sub>, O<sub>3</sub>, SO<sub>2</sub>, OH, isoprene, and monoterpene), and meteorological variables (T, RH, precipitation, and solar radiation)
- Also captures [CCN0.4] variability & magnitude at ARM SGP

## Geophysical Research Letters Machine Learning Uncovers Aerosol Size Information From Chemistry and Meteorology to Quantify Potential Cloud-Forming Particles

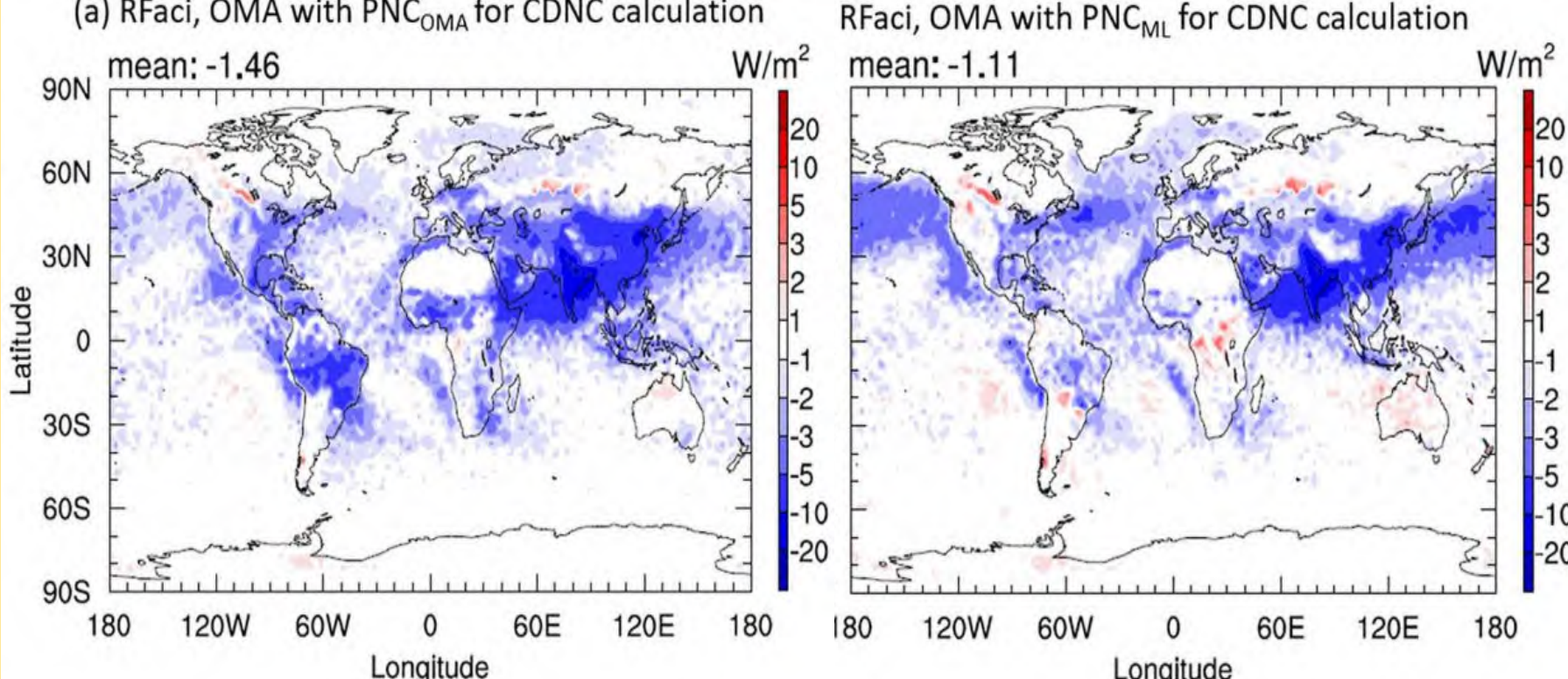
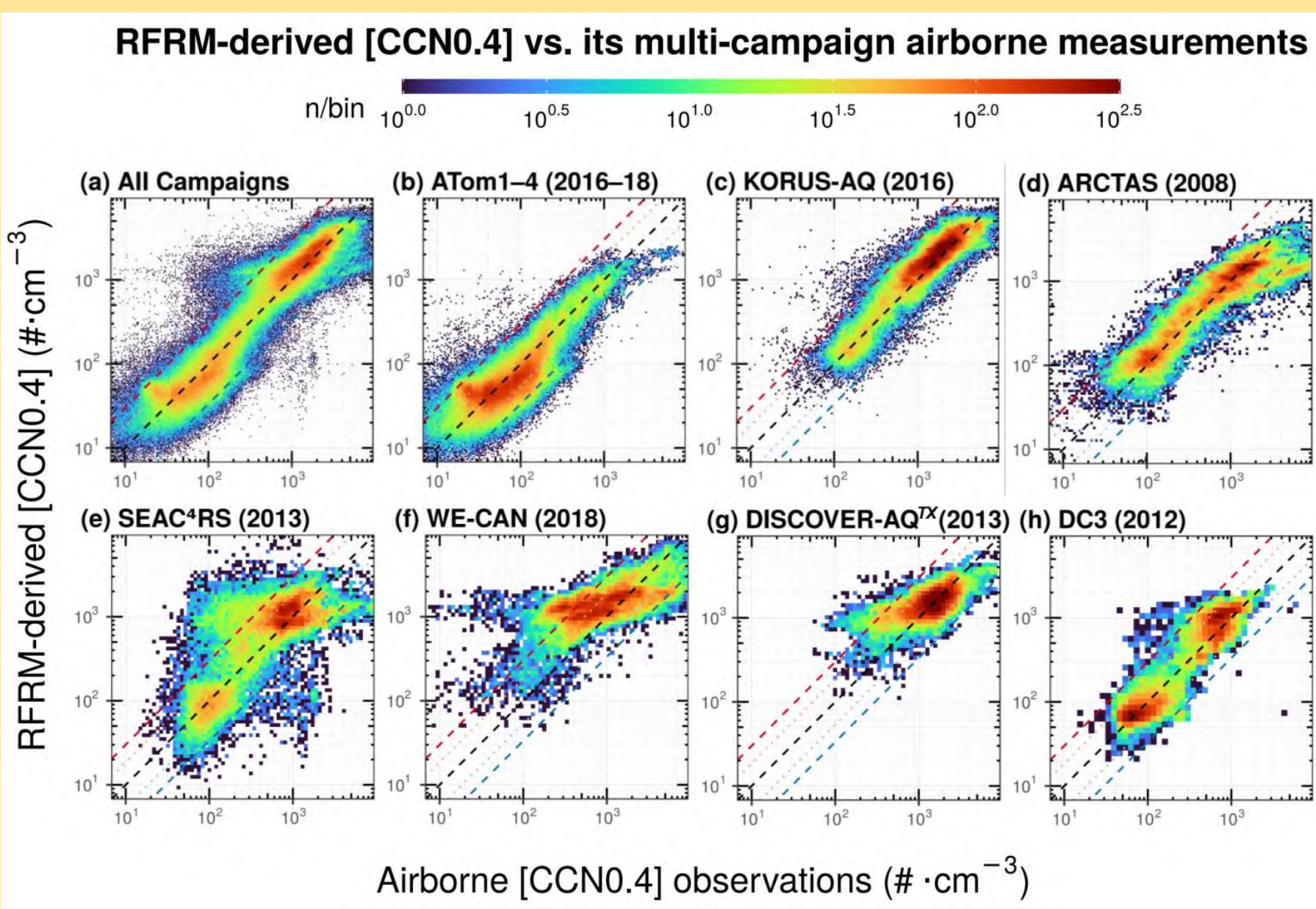
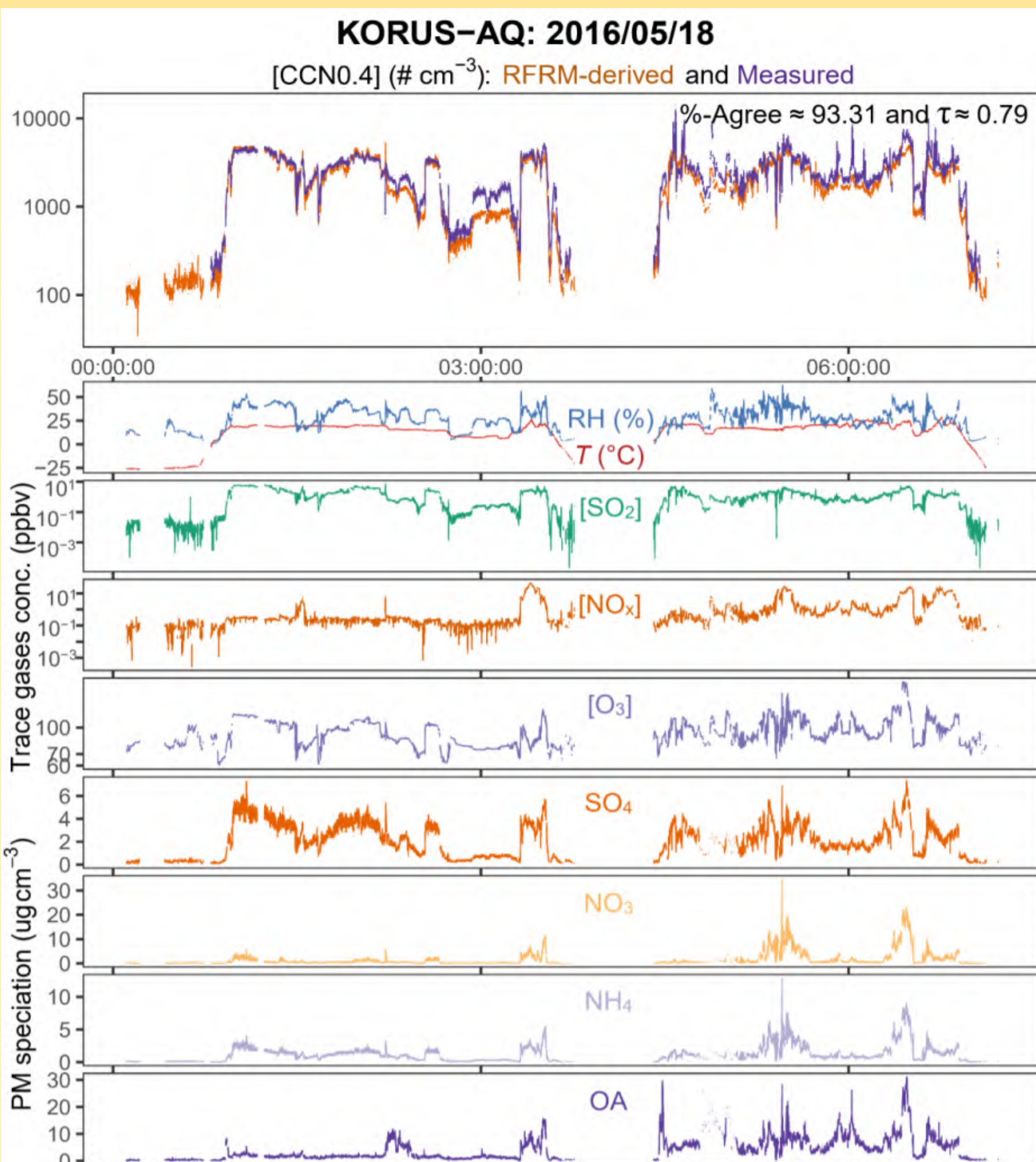
Arshad Arjunan Nair<sup>1</sup>, Fangqun Yu<sup>1</sup>, Pedro Campuzano-Jost<sup>2,3</sup>, Paul J. DeMott<sup>4</sup>, Ezra J. T. Levin<sup>4,5</sup>, Jose L. Jimenez<sup>2,3</sup>, Jeff Peischl<sup>2,6</sup>, Ilana B. Pollack<sup>4</sup>, Carley D. Fredrickson<sup>7</sup>, Andreas J. Beyersdorf<sup>9</sup>, Benjamin A. Nault<sup>2,3,10</sup>, Minsu Park<sup>11</sup>, Seong Soo Yum<sup>11</sup>, Brett B. Palm<sup>7</sup>, Lu Xu<sup>12,13</sup>, Ilann Bourgeois<sup>2,6</sup>, Bruce E. Anderson<sup>8</sup>, Athanasios Nenes<sup>14,15,16</sup>, Luke D. Ziemba<sup>8</sup>, Richard H. Moore<sup>8</sup>, Taehyoung Lee<sup>17</sup>, Taehyun Park<sup>17</sup>, Chelsea R. Thompson<sup>2,6</sup>, Frank Flocke<sup>18</sup>, Lewis Gregory Huey<sup>19</sup>, Michelle J. Kim<sup>12</sup>, and Qiaoyun Peng<sup>7</sup>

- ML-derived CCN numbers in strong agreement with multi-campaign aircraft measurements
- Aerosol size information is contained in speciated aerosol mass, chemistry, and meteorology and is extractable by ML

## Geophysical Research Letters Use of Machine Learning to Reduce Uncertainties in Particle Number Concentration and Aerosol Indirect Radiative Forcing Predicted by Climate Models

Fangqun Yu<sup>1</sup>, Gan Luo<sup>1</sup>, Arshad Arjunan Nair<sup>1</sup>, Kostas Tsigaridis<sup>2,3</sup>, and Susanne E. Bauer<sup>2</sup>

- Trained using GEOS-Chem-APM, the ML model adds only ~3.1% overhead to GEOS-Chem Classic
- Also implemented in GISS-ModelE2.1-OMA with PNC now agreeing better with measurements



- RF<sub>aci</sub> changes from -1.46 to -1.11 W·m<sup>-2</sup>; closer to median IPCC value and GISS-ModelE2.1-MATRIX
- Highlights need to account for the particle size changes from PI to PD in PNC and CDNC calculations

## GEOS-Chem-APM for pollutant exposure & health impacts

Environmental exposure disparities in ultrafine particles and PM<sub>2.5</sub> by urbanicity and socio-demographics in New York state, 2013–2020

Arshad Arjunan Nair<sup>a,\*</sup>, Shao Lin<sup>b,c</sup>, Gan Luo<sup>a</sup>, Ian Ryan<sup>c</sup>, Quan Qi<sup>d</sup>, Xinlei Deng<sup>c</sup>, Fangqun Yu<sup>a,\*</sup>

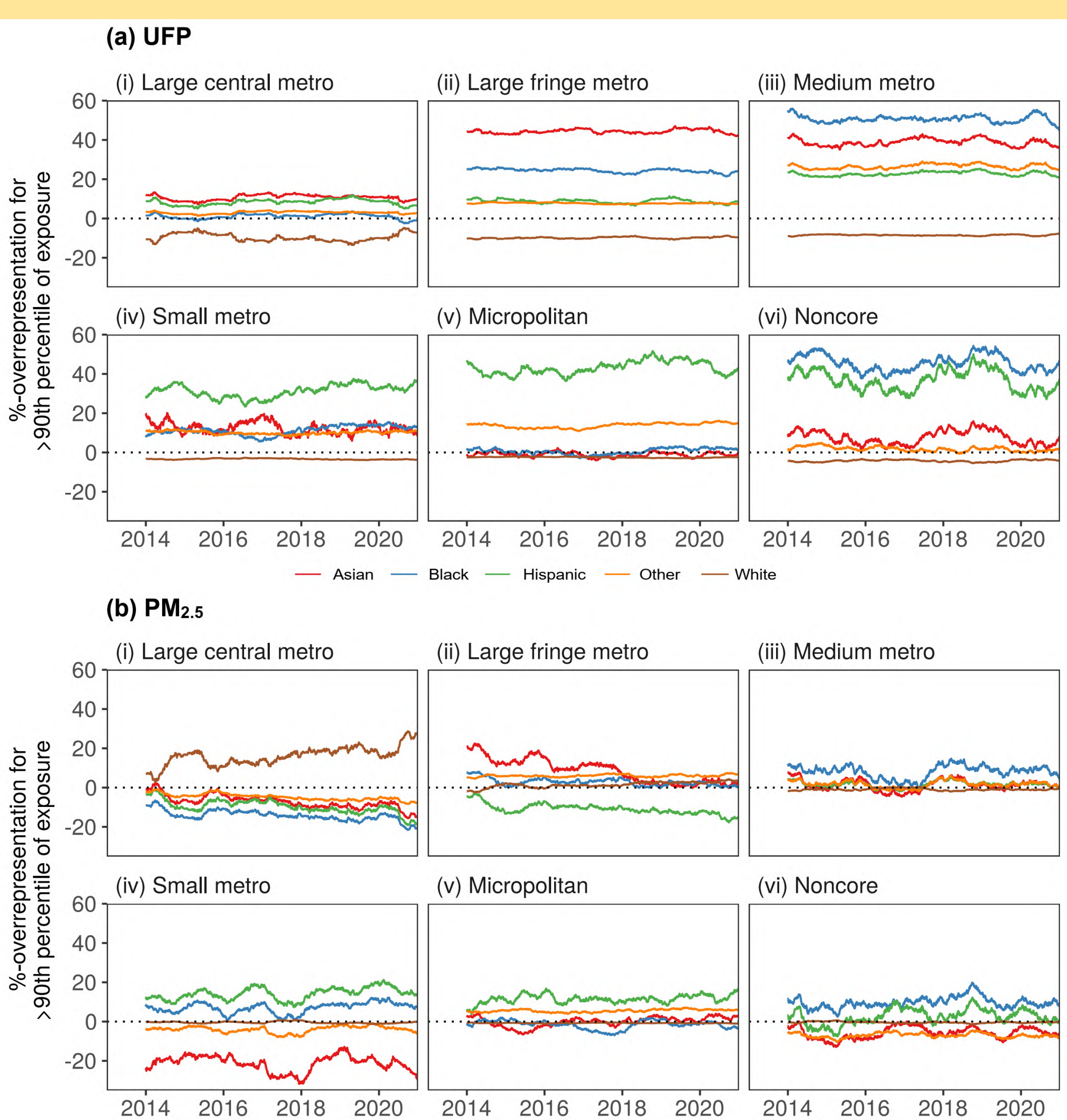
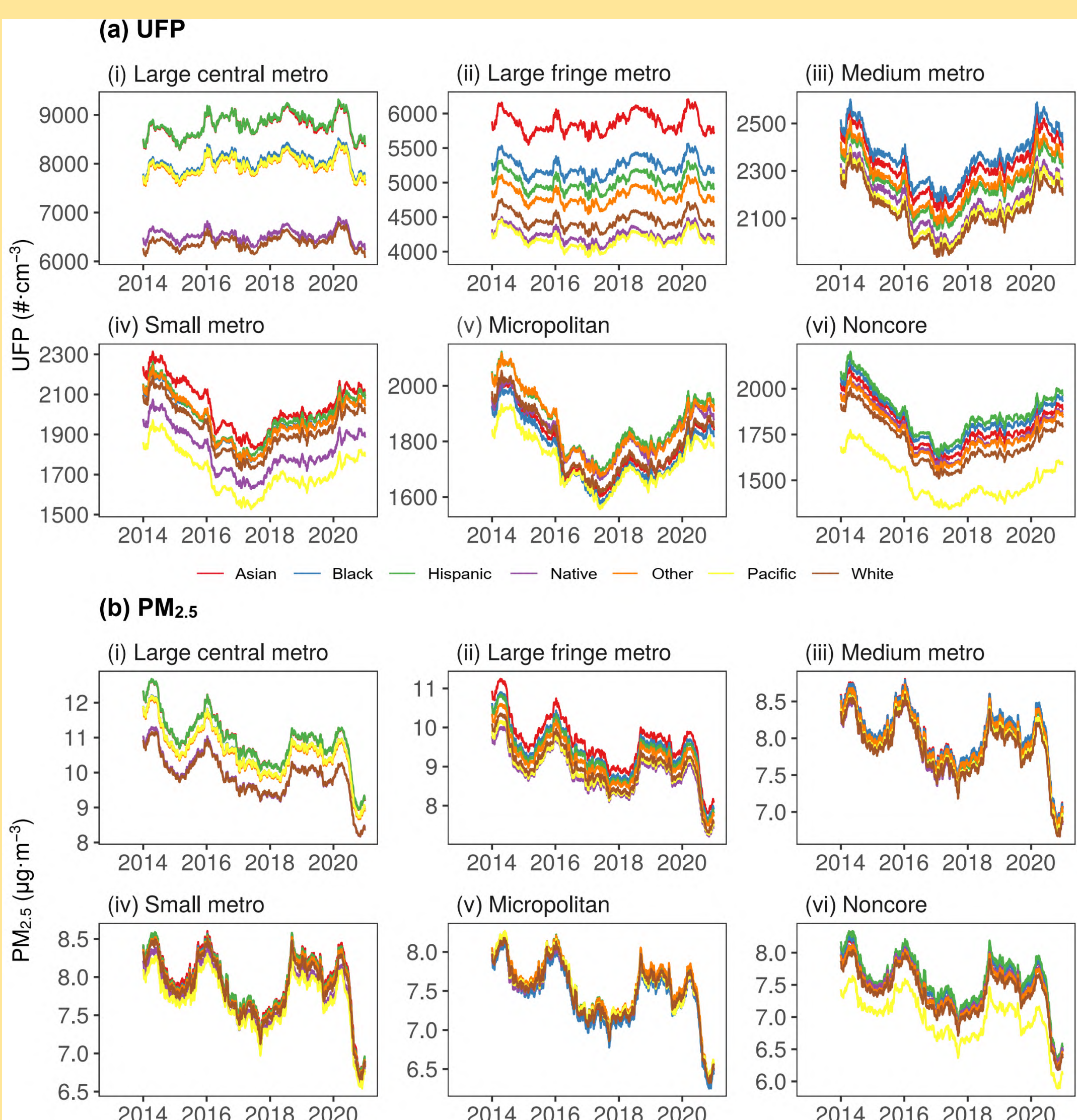
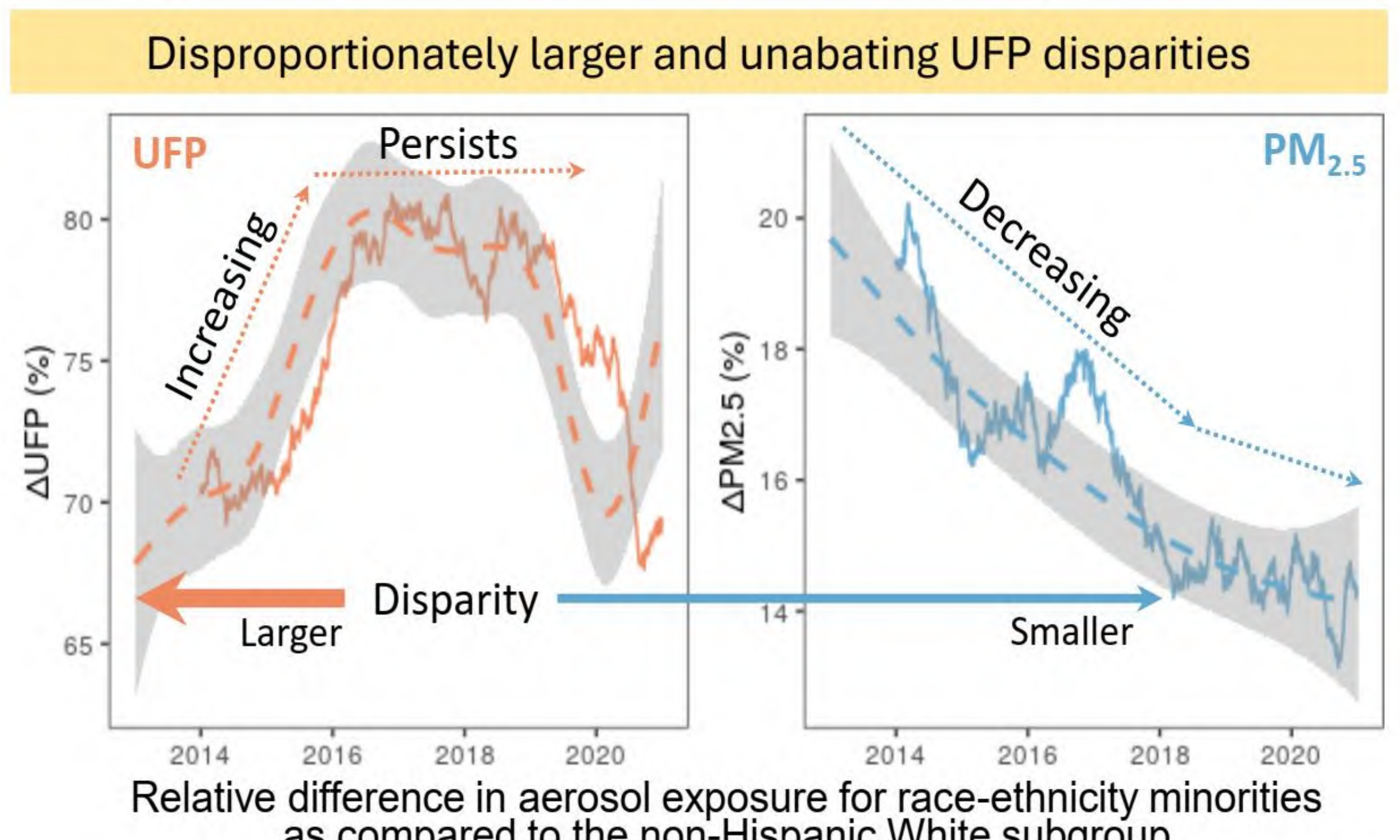
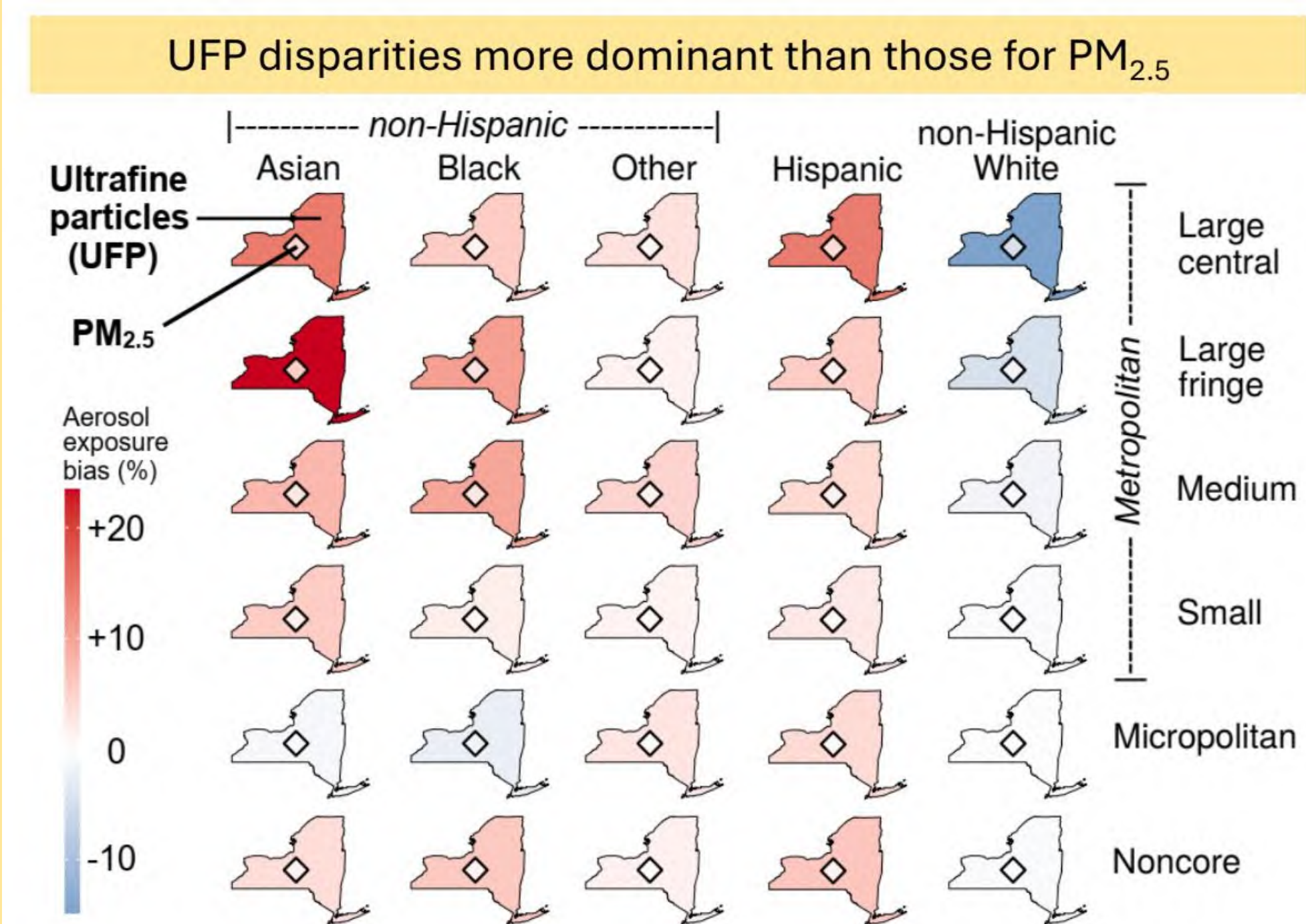


Figure. Population-weighted exposure at the county subdivision level in NYS during 2013–2020.

Figure. Race-ethnicity representation bias in the worst 10% exposure jurisdictions.

